

Working with (Illumina) Sequences

- Postponing Homework Assignment #1 is until next Tue at 11:59pm
 - Let me or Sonali know if you need after hours access to the computer labs to work on assignments
- Open RStudio
- Download from OneDrive 03_Working_with_Sequences.Rmd
- run the first two chunks:
 - 1) `{r setup, include=FALSE}`
 - 2) `{r install-packages}`

A “standard” Illumina sequencing workflow

- 1) obtain nucleic acids (DNA or RNA)
- 2) generate Illumina library
 - fragment nucleic acids (~300-bp)
 - ligate adapters (or transposon “tagmentation,” which both fragments and ligates adapters at the same time)
 - Note: can use “barcoded” adapters where each sample receives a unique known DNA barcode for multiplexing
 - PCR amplify library
- 3) sequence on flowcell (performed at dedicated sequencing centers)
- 4) receive millions or reads and their quality scores per nucleotide for each sample

A “standard” Illumina sequencing workflow

- 5) perform quality control analysis of each sample
- 6) align reads to a known reference genome (or transcriptome), or assemble the genome de novo (if it's completely new)
- 7) assign reads to genes or generate annotations de novo to identify genes in your new genome
- 8) perform downstream analysis based on why you sequenced the samples
 - e.g., for RNA-seq, determine the number of read counts that map to each gene and see how those differ between samples
 - e.g., for DNA, identify allelic differences between samples to identify polymorphisms

Standard Illumina RNA-seq Library Construction

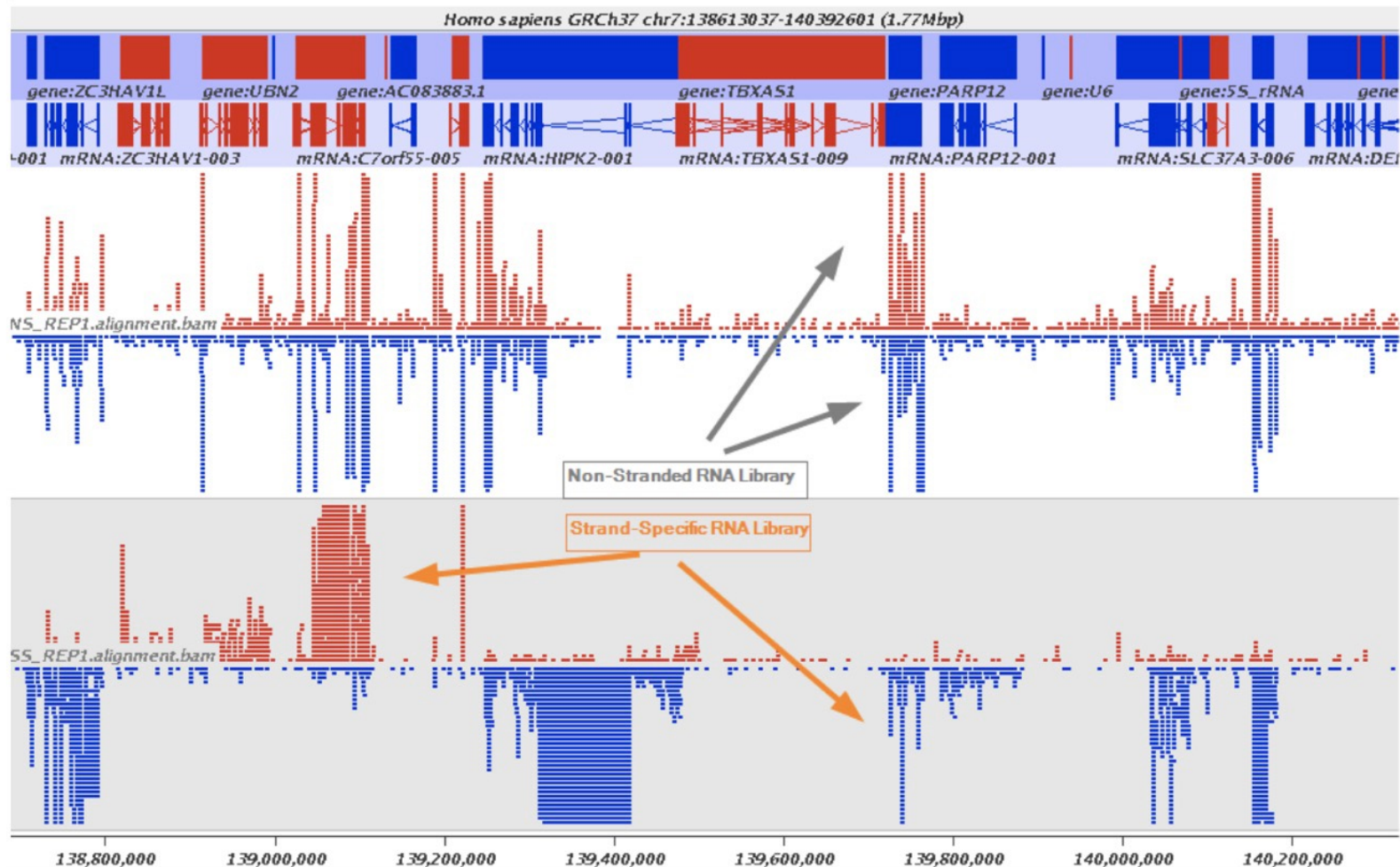
- 1) Isolate high quality total RNA from cells
 - phenol/chloroform extraction, kits, etc.
- 2) Enrich for mRNA
 - rRNA depletion (use beads conjugated with oligos that are complementary to rRNA)
 - poly-A enrichment (use beads conjugated with poly-T oligos)
- 3) Fragment mRNA (heat plus magnesium buffer)
 - fragments should be longer than the read length (e.g., 350-nt fragments for paired-end 100-bp reads (200-bp plus 2 x 70-nt adapters))
- 4) 1st strand cDNA synthesis using random hexamer priming and reverse transcriptase

Standard Illumina RNA-seq Library Construction

- 5) Second strand synthesis and “marking” for strand-specific RNA-seq
 - dUTP incorporated into second strand instead of dTTP, yielding double-stranded (ds) DNA

Standard Illumina RNA-seq Library Construction

The implication of stranded RNAseq is that you can distinguish whether the reads are derived from forward- or reverse-encoded transcripts:

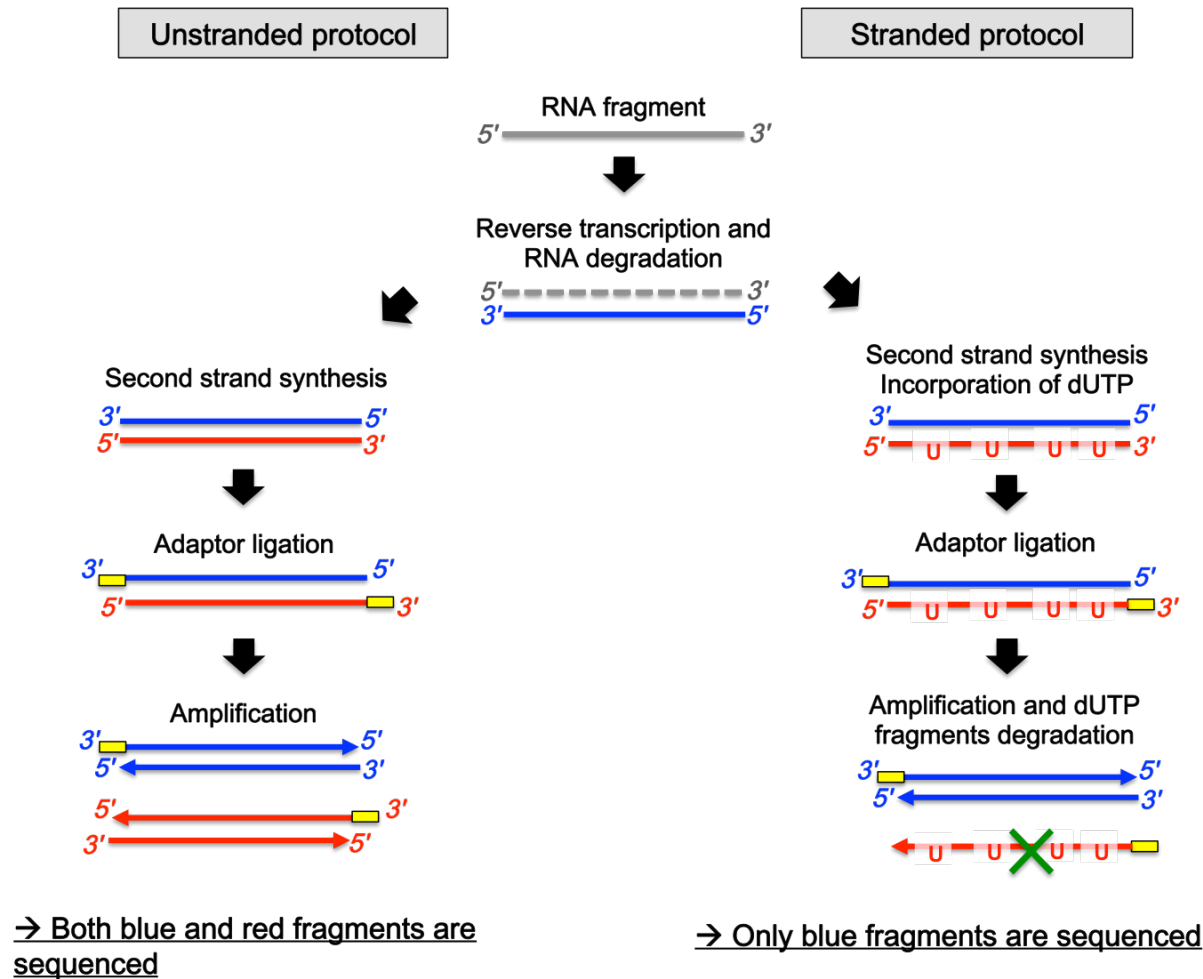


Stranded RNAseq data look like this

This example contrasts unstranded and stranded RNAseq experiments. Red transcripts are from + strand and blue are from - strand. In stranded example reads are clearly stratified between the two strands. A small number of reads from opposite strand may represent anti-sense transcription. The image from GATC Biotech.

Standard Illumina RNA-seq Library Construction

- 5) Second strand synthesis and “marking” for strand-specific RNA-seq
- dUTP incorporated into second strand instead of dTTP, yielding double-stranded (ds) DNA



Standard Illumina RNA-seq Library Construction

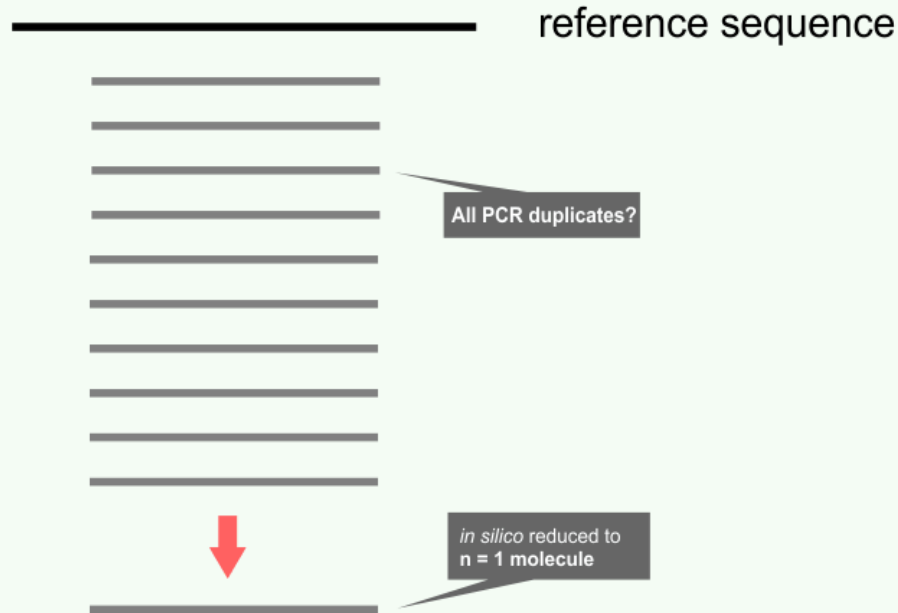
- 5) Second strand synthesis and “marking” for strand-specific RNA-seq
 - dUTP incorporated into second strand instead of dTTP, yielding double-stranded (ds) DNA
- 6) A-tailing, which adds dAMP (“A”) to the 3’ ends of the dsDNA
- 7) Adapter ligation, where adapters have 3’ “T” overhangs that allow complementary binding to the “A” overhangs of the dsDNA.
- 8) PCR amplify library with primers complementary to the adapter sequence
 - DNA pol “stops” at UTP, so those strands are not amplified
 - primers can contain unique DNA barcodes for multiplexing (e.g., sample #1 gets TATAGCCT, sample #2 gets CGAATATC)
 - primers can also have unique molecular identifiers (UMI), which are random barcodes instead of fixed sequences

UMIs

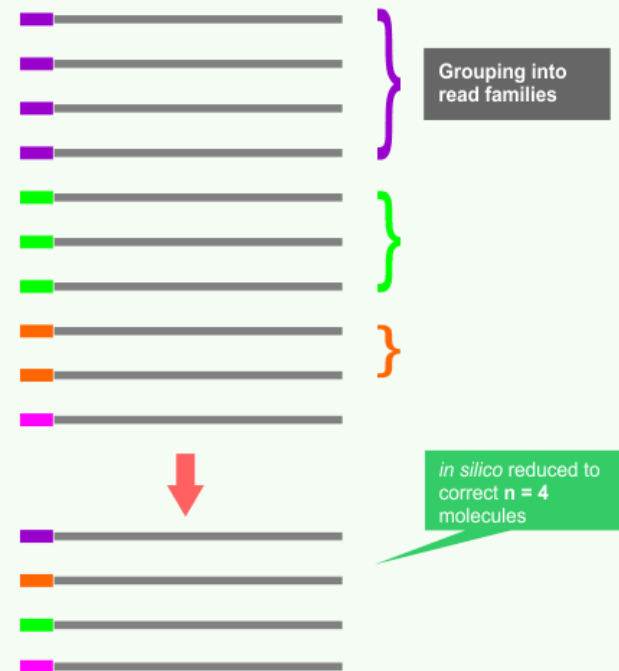
Confident analysis of reads sharing the same alignment coordinates.

UMI application in **quantitative studies** (e.g. RNA-seq, scRNA-seq, miRNA-Seq, ChIP-seq).

PCR duplicate removal without UMIs



PCR duplicate removal with UMIs



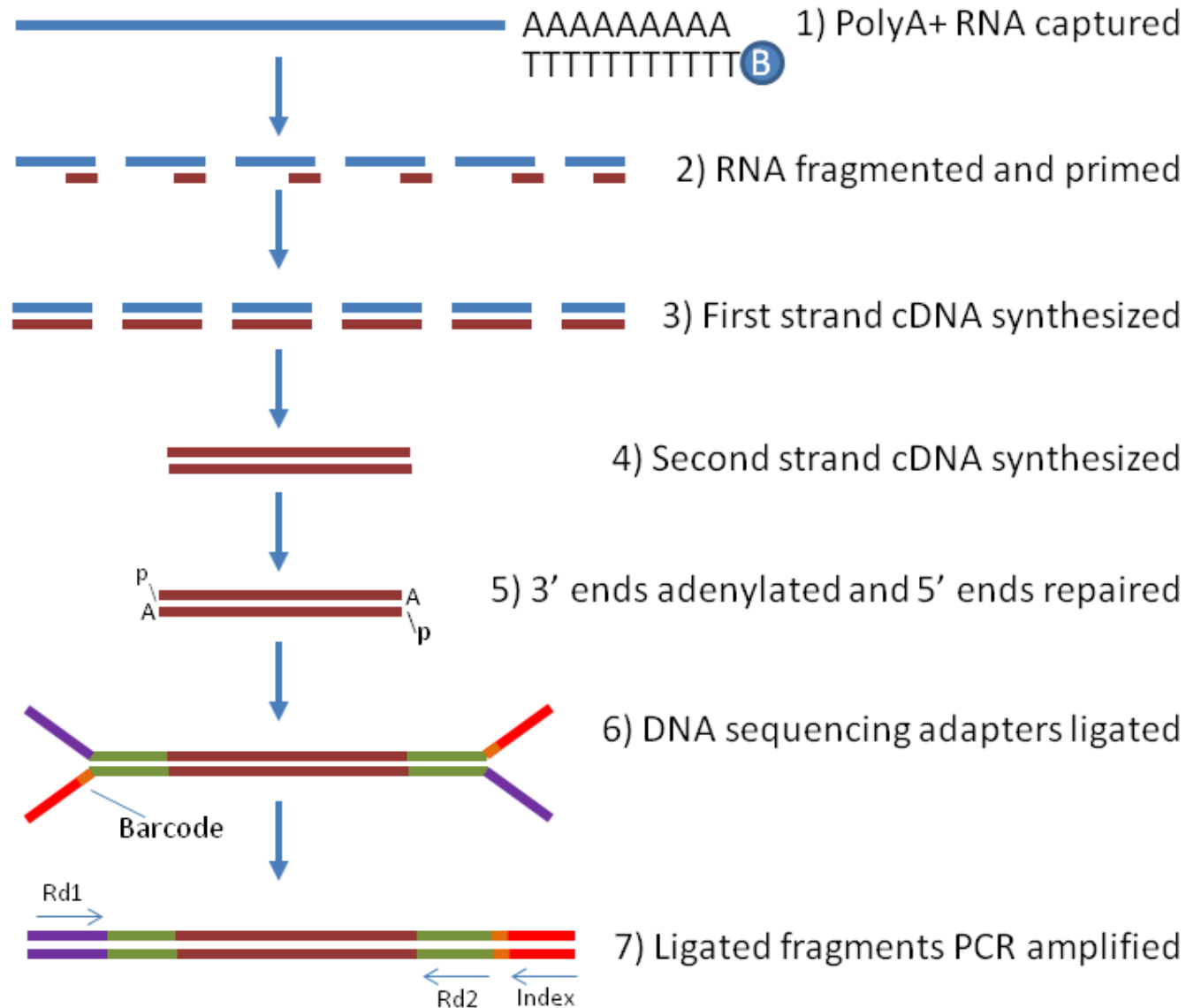
Recommended for very low input samples (single cell RNA-seq, ChIP-seq), very deep sequencing (>80M reads/sample), detection of ultra-low frequency mutations

<https://dnatech.genomecenter.ucdavis.edu/faqs/what-are-umis-and-why-are-they-used-in-high-throughput-sequencing/>

Standard Illumina RNA-seq Library Construction

- 9) Cleanup and size selection to remove unligated adapters and primers
- 10) Library quality control (QC)
 - make sure quantity is sufficient for sequencing
 - electrophoresis to make sure fragment size is correct

Standard Illumina RNA-seq Library Construction



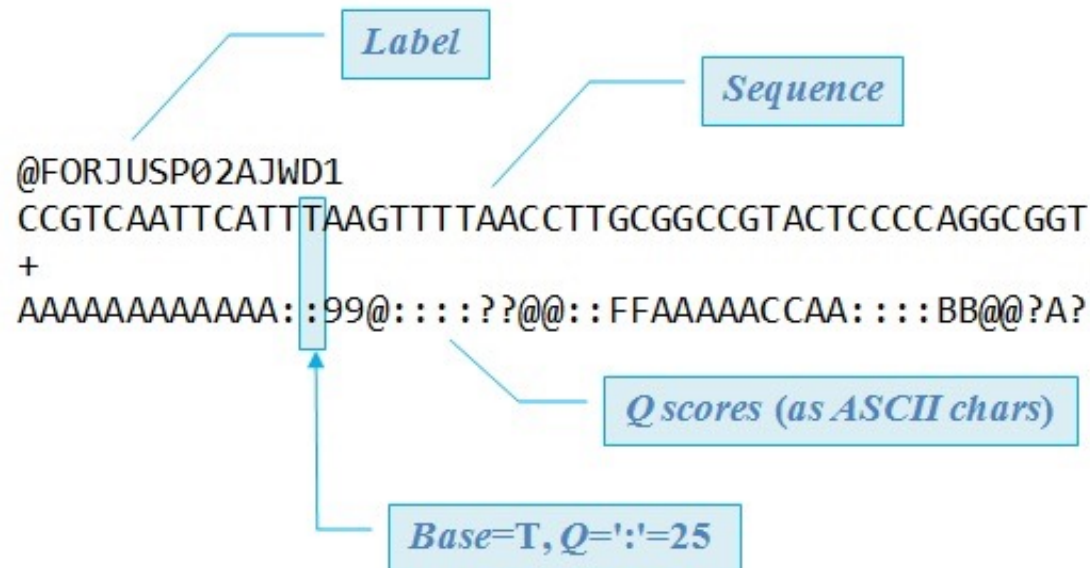
The FASTQ File

- following Illumina sequencing, basecalls are generated from the raw images along with quality scores, which are all added to a FASTQ file for each sample
- the FASTQ format is a text format that stores nucleotide sequences and corresponding quality scores
- Is an extension of the FASTA format, that just stores sequences (and not quality scores)
- FASTQ files are compressed in GNU zip and end with the .gz file extension

The FASTQ File

- Each FASTQ entry consists of 4 lines:
 - sequence identifier (unique name)
 - the nucleotide sequence
 - quality score identifier line (consisting only of a +)
 - quality scores as ASCII characters (with each character designating a score ranging from 1 – 40)

one line of FASTQ sequence



Quality scores

Symbol	ASCII Code	Q-Score	Symbol	ASCII Code	Q-Score
!	33	0	6	54	21
"	34	1	7	55	22
#	35	2	8	56	23
\$	36	3	9	57	24
%	37	4	:	58	25
&	38	5	;	59	26
'	39	6	<	60	27
(	40	7	=	61	28
)	41	8	>	62	29
*	42	9	?	63	30
+	43	10	@	64	31
,	44	11	A	65	32
-	45	12	B	66	33
.	46	13	C	67	34
/	47	14	D	68	35
0	48	15	E	69	36
1	49	16	F	70	37
2	50	17	G	71	38
3	51	18	H	72	39
4	52	19	I	73	40
5	53	20			

Quality scores

Phred scores: a negative log scale (so a higher Phred score means more accurate)

Phred Quality Score	Probability of incorrect base call	Base call accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1000	99.9%
40	1 in 10000	99.99%
50	1 in 100000	99.999%

How to find publicly available sequences

Sequence Read Archive (SRA)

Gene Expression Omnibus (GEO)

SRA

- public repository for DNA sequencing data
- established by the National Center for Biotechnology Information (NCBI) in 2007 as the Short Read Archive, but name was changed to reflect the growth of long-read sequencing
- currently supported by the National Center for Biotechnology Information (NCBI, part of the NIH), the European Bioinformatics Institute (EBI), and the DNA Data Bank of Japan (DDBJ).

GEO

- more specific public repository for gene expression data (microarrays, RNA-seq, ChIP-seq, etc.)
- created in 2002 by NCBI
- sequences deposited onto GEO are automatically also deposited onto SRA
- GEO datasets contain metadata for how the experiments were performed

How to access data

Manually download from GEO or SRA (either web based or through an FTP server)

Use the SRA Toolkit to download:

<https://www.ncbi.nlm.nih.gov/sra/docs/srdownload/>

Use the R package GEOquery:

<https://bioconductor.org/packages/release/bioc/html/GEOquery.html>

How do you know that your data downloaded fully?

MD5 checksums:

The MD5 algorithm takes a data file and generates a unique 32-character hexadecimal number (e.g., the string “frog” always generates 938c2cc0dcc05f2b68c4287040cfcf71)

If the data is changed, the MD5 output will be different.

So, if you download a file and the MD5 checksum matches that of the file that was uploaded, you can be confident that the file is an exact copy.

To obtain the checksum in Terminal, type `md5 <file>`

Downloads for Thursday

Bioconda (for Mac or Linux machines):

<https://bioconda.github.io/>

FastQC

<https://www.bioinformatics.babraham.ac.uk/projects/download.html#fastqc>

MultiQC

https://multiqc.info/docs/getting_started/installation/